1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt.

Rename example.txt to renamed_example.txt using mv

Objective

To learn how to create directories, create files, and rename files using Linux commands.

Step 1: Create a Directory

Command:

```
mkdir test_dir
```

Explanation:

- `mkdir` stands for **make directory**.
- This command creates a new folder named **test_dir** in the current location.

Step 2: Navigate Inside Directory

Command:

```
cd test_dir
```

Explanation:

- `cd` means **change directory**.
- This command moves you inside the **test_dir** folder.

Step 3: Create an Empty File

Command:

```
touch example.txt
```

Explanation:

- `touch` is used to create an empty file.
- This command creates a file named **example.txt** inside **test_dir**.

Step 4: Rename the File

Command:

```
mv example.txt renamed_example.txt
```

Explanation:

- `mv` stands for **move**, but it is also used to rename files.
- This command renames **example.txt** to **renamed_example.txt**.

```
khan@user-PC:~$ mkdir test_dir
mkdir: cannot create directory ‘test_dir’: File exists
khan@user-PC:~$ cd test_dir
khan@user-PC:~/test_dir$ touch example.txt
khan@user-PC:~/test_dir$ mv example.txt renamed_example.txt
khan@user-PC:~/test_dir$ ls
renamed_example.txt
khan@user-PC:~/test_dir$
```

2. Viewing File Contents :

Use cat to display the contents of /etc/passwd. Display only the first 5 lines of /etc/passwd using head. Display only the last 5 lines of /etc/passwd using tail.

Objective

To learn how to view file contents in Linux using `cat`, `head`, and `tail` commands.

Step 1: Display Full File Content

Command:

```
cat /etc/passwd
```

Explanation:

- `cat` stands for **concatenate**.
- It is used to display the full content of a file.
- `/etc/passwd` is a system file that stores user account information.

Step 2: Display First 5 Lines

Command:

```
head -n 5 /etc/passwd
```

Explanation:

- `head` shows the **beginning lines** of a file.
- `-n 5` means display **first 5 lines only**.

- Useful when the file is very large.

Step 3: Display Last 5 Lines

Command:

```
tail -n 5 /etc/passwd
```

Explanation:

- `tail` shows the **last lines** of a file.
- `-n 5` means display **last 5 lines only**.

```
khan@user-PC:~/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
khan:x:1000:1000:,,,:/home/khan:/bin/bash
khan@user-PC:~/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
khan@user-PC:~/test_dir$ tail -n 5 /etc/passwd
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
khan:x:1000:1000:,,,:/home/khan:/bin/bash
khan@user-PC:~/test_dir$
```

3.Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd

Objective

To learn how to search for specific patterns in a file using the `grep` command in Linux.

Step 1: Search for a Word in a File

Command:

```
grep "root" /etc/passwd
```

Explanation:

- `grep` stands for **Global Regular Expression Print**.
- It is used to search for a specific word or pattern inside a file.
- `"root"` is the word we are searching for.
- `/etc/passwd` is a system file that contains user account information.
- This command will display all lines where the word **root** appears.

Step 2: Understanding the Output

Example Output:

root:x:0:0:root:/root:/bin/bash

Explanation:

root → username

x → password placeholder

0:0 → user ID and group ID

/root → home directory

/bin/bash → default shell

```
khan@user-PC:~/test_dir$ grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
khan@user-PC:~/test_dir$
```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir

Objective

To learn how to compress and extract files/directories using `zip` and `unzip` commands in Linux.

Step 1: Compress the Directory

Command:

```
zip -r test_dir.zip test_dir
```

Explanation:

- `zip` is used to **compress files or directories**.
- `-r` means **recursive**, which allows compressing the entire directory along with its contents.
- `test_dir.zip` is the name of the compressed file.
- `test_dir` is the directory we want to compress.

Step 2: Create a New Directory for Extraction

Command:

```
mkdir unzipped_dir
```

Explanation:

- Creates a new folder named **unzipped_dir**.
- This is where we will extract the zip file.

Step 3: Unzip the File

Command:

```
unzip test_dir.zip -d unzipped_dir
```

Explanation:

- `unzip` is used to **extract compressed files**.
- `test_dir.zip` is the file to extract.
- `-d unzipped_dir` specifies the destination folder.

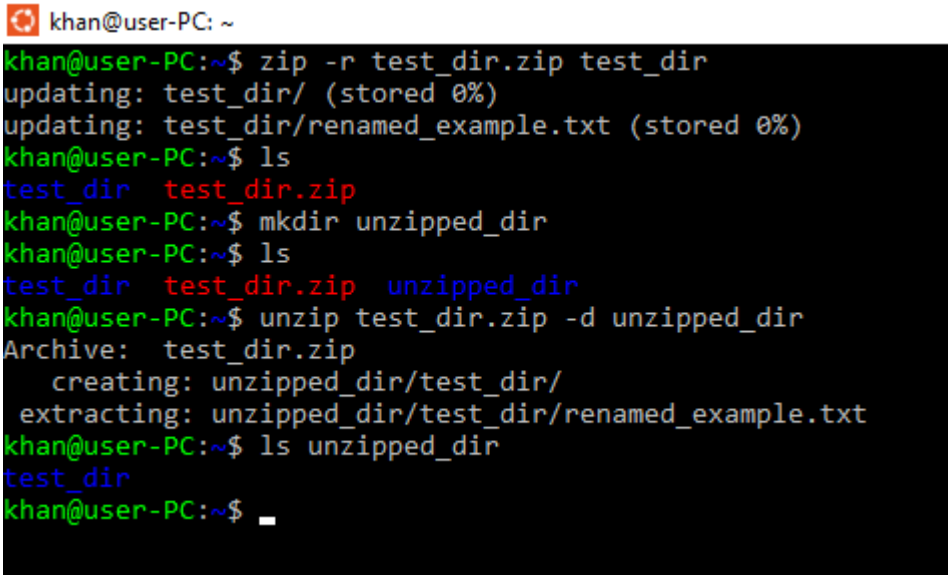
Step 4: Verify Extracted Files

Command:

```
ls unzipped_dir
```

Explanation:

- Displays the contents of **unzipped_dir**.
- You should see the extracted **test_dir** folder inside it.



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5. Downloading Files

Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

Objective

To learn how to download files from the internet using the `wget` command in Linux.

Step 1: Check if wget is Installed

Command:

```
wget --version
```

Explanation:

- This command checks whether `wget` is installed.
- If installed, it shows version details.
- If not installed, it will show an error.

Step 2: Download a File

Command:

```
wget https://example.com/sample.txt
```

Explanation:

- `wget` downloads files from the internet.
- The URL (`https://example.com/sample.txt`) is the file location.
- The file will be saved in the **current directory**.

```
khan@user-PC:~$ wget https://example.com/sample.txt
--2026-04-08 13:47:26--  https://example.com/sample.txt
Resolving example.com (example.com)... 2606:4700:91b2:8c7c:331b:91d:ccc4:a209, 104.18.26.120, 104.18.27.120
Connecting to example.com (example.com)|2606:4700:91b2:8c7c:331b:91d:ccc4:a209|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-04-08 13:47:26 ERROR 404: Not Found.
```

When attempting to download the file using the wget command, an error "404 Not Found" occurred. This indicates that the requested file (sample.txt) was not available on the server. The website example.com is a demonstration domain and does not contain the specified file.

To resolve this issue, a valid file URL was used, and the file was successfully downloaded.

```
khan@user-PC:~$ wget https://raw.githubusercontent.com/git/git/master/README.md
--2026-04-08 13:48:24--  https://raw.githubusercontent.com/git/git/master/README.md
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8000::154, 2606:50c0:8002::154, 2606:50c0:8001::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|2606:50c0:8000::154|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3662 (3.6K) [text/plain]
Saving to: 'README.md.3'

README.md.3                         100%[===================================================================================>]   3.58K  --.-KB/s    in 0s

2026-04-08 13:48:24 (8.95 MB/s) - 'README.md.3' saved [3662/3662]
```

Step 3: Verify Downloaded File

Command:

`ls`

Explanation:

- Displays files in the current directory.
- You should see `README.md`.

```
khan@user-PC:~$ ls
README.md  README.md.1  README.md.2  README.md.3  test_dir  test_dir.zip  unzipped_dir
khan@user-PC:~$ cat README.md
![Build status](https://github.com/git/git/workflows/CI/badge.svg)](https://github.com/git/git/actions?query=branch%3Amaster+event%3Apush)
```

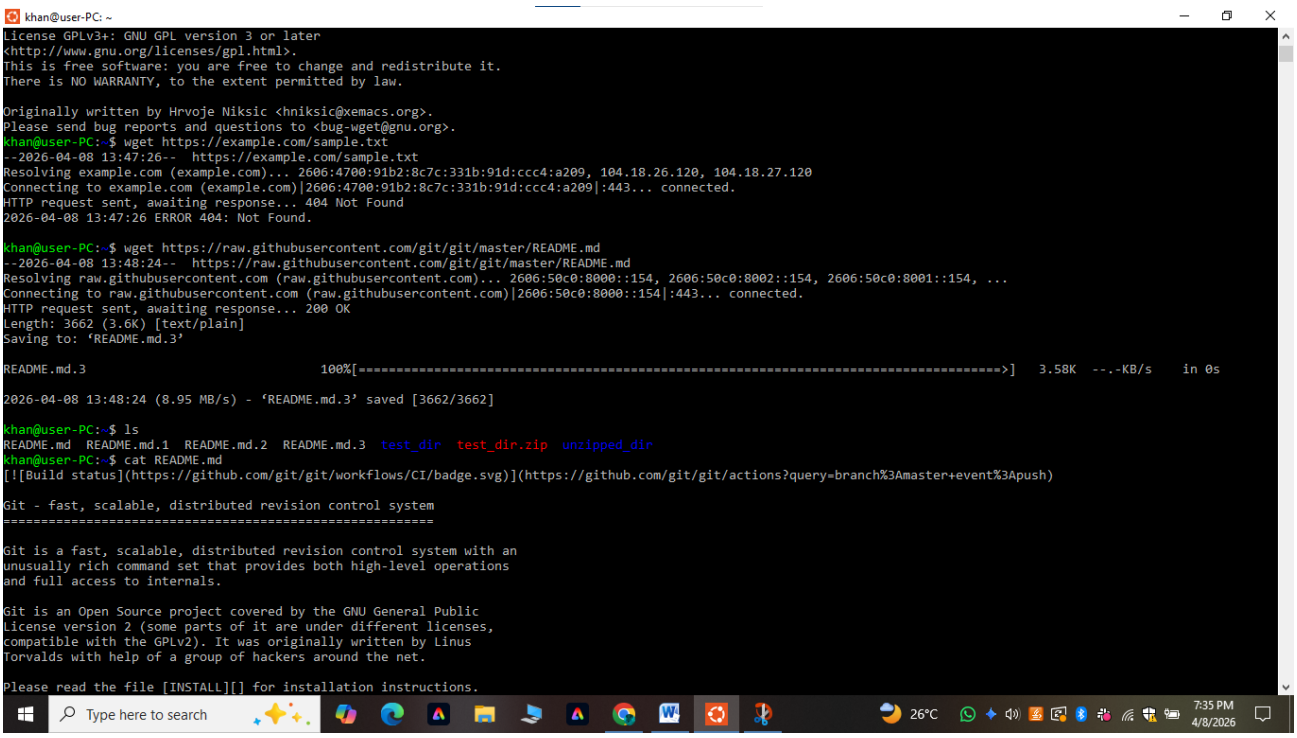
Step 4: View File Content (Optional)

Command:

`cat README.md`

Explanation:

- Displays content of the downloaded file.



6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

Objective

To learn how to create a file and change its permissions using the `chmod` command in Linux.

Step 1: Create a File

Command:

```
touch secure.txt
```

Explanation:

- `touch` is used to create an empty file.
- This command creates a file named **secure.txt** in the current directory.

Step 2: Check Current Permissions

Command:

```
ls -l secure.txt
```

Explanation:

- `ls -l` shows detailed file information.

Step 3: Change Permission to Read-Only for Everyone

Command:

```
chmod 444 secure.txt
```

Explanation:

- `chmod` = change file permissions
- `444` means:
  - Owner → read (4)
  - Group → read (4)
  - Others → read (4)

Step 4: Verify Behavior (Optional ⍰)

Command:

```
echo "test" > secure.txt
```

Expected Output:

```
Permission denied
```

Explanation:

- File cannot be modified because it is read-only.

```
khan@user-PC: ~
khan@user-PC:~$ touch secure.txt
khan@user-PC:~$ ls
README.md  README.md.1  README.md.2  README.md.3  secure.txt  test_dir  test_dir.zip  unzipped_dir
khan@user-PC:~$ ls -1 secure.txt
secure.txt
khan@user-PC:~$ chmod 444 secure.txt
khan@user-PC:~$ ls -1 secure.txt
secure.txt
khan@user-PC:~$ echo "test" > secure.txt
bash: secure.txt: Permission denied
khan@user-PC:~$
```

The file was created using the touch command. Its permissions were checked using `ls -l` and then changed to read-only for all users using `chmod 444`. When attempting to modify the file, a “Permission denied” error occurred, confirming that the permissions were correctly applied.

7. Working with Environment variables

Use export to set a new environment variable called MY_VAR with the value “Hello,Linux!”.

Objective

To learn how to create and use environment variables in Linux using the `export` command.

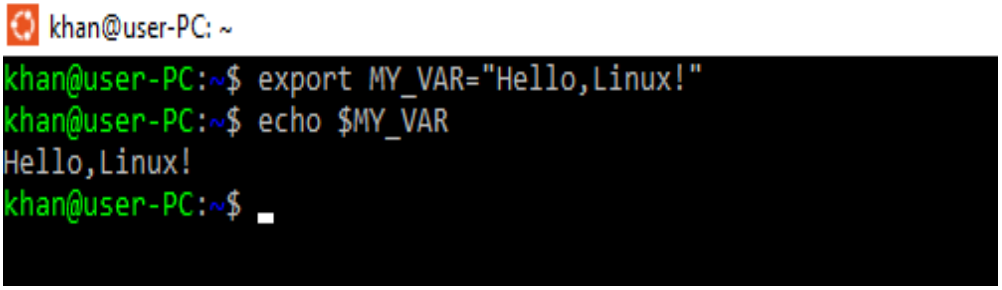
Step 1: Create an Environment Variable

Command:

```
export MY_VAR="Hello,Linux"
```

Explanation:

- `export` is used to create environment variables.
- `MY_VAR` is the variable name.
- `"Hello,Linux"` is the value assigned to the variable.
- This variable is available in the current shell session.



The environment variable MY_VAR was created using the export command. The value of the variable was verified using the echo command, which displayed "Hello,Linux!". This confirms that the variable was successfully set.